

RICARDO HOAR

Engineering Leader | Developer Advocate | Technical Author

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Summary

Engineering leader and technical communicator with 20+ years building high-performance teams, delivering innovative products, and bridging technical and business stakeholders. Proven track record in game development automation, mobile platforms, curriculum design, and developer advocacy. Co-author of industry-standard web development textbook adopted by 200+ institutions globally.

Core Competencies

- Engineering Leadership & Team Building
- Developer Relations & Advocacy
- Test Automation & CI/CD
- Game Development (Unreal/Unity)
- Mobile Development (iOS/Android)
- Technical Writing & Documentation
- Curriculum Development
- Agile Methodologies

Professional Experience

Development Manager

Confidential Tech Company

Jun 2022 – Present

Remote

- Architected cross-platform test automation framework for **Unreal Engine 4.27-5.6** (C++/C#), enabling QA teams to automate game testing at scale across multiple studio titles
- Pioneered **MCP (Model Context Protocol) AI server** integration, exposing company APIs and Unreal/Unity games to LLMs like Claude for AI-assisted test development
- Led distributed team of 5 engineers using agile methodologies, delivering quarterly releases with 95%+ on-time delivery rate
- Created comprehensive video tutorials, quickstart guides, and sample applications, reducing developer onboarding time by 60%
- Served as internal developer advocate, partnering with Product and Engineering to prioritize features based on community feedback and telemetry data

President, CEO

Silicon Hanna Inc

Jun 2005 – Present

Remote

- Co-founded boutique software development studio with focus on iOS app development, managing all aspects of product strategy, development, and operations
- Shipped 12+ mobile titles across Android/iOS platforms using **Unreal Engine** with native SDK integration (Swift/Kotlin), generating 500k+ downloads
- Engineered real-time multiplayer systems and puzzle games with Apple API integrations (GameCenter, ARKit)

- Evolved product focus from casual games to educational AR applications, expanding into EdTech market

Associate Dean

Sheridan College, School of Applied Computing

Jan 2017 – Feb 2018

Ontario

- Directed academic operations for 11 STEM programs serving 2,000+ students across diplomas, degrees, and post-graduate certificates
- Led \$1.2M industry partnership initiatives for applied VR/AR research, collaborating with major tech companies
- Managed 25+ full-time faculty and 100+ part-time instructors, delivering 200+ technical courses annually
- Orchestrated institutional visit by Apple CEO Tim Cook, coordinating technical demonstrations and community engagement
- Championed diversity initiatives including Women in Computing program, increasing female enrollment by 30%
- Represented institution at technical conferences and industry events, building strategic partnerships with Apple, Microsoft, and Unity

Lead Technical Author

Pearson Education

Mar 2014 – Sep 2021

Global

- Co-authored 3 editions of **"Fundamentals of Web Development"**, industry-standard textbook adopted by 200+ institutions worldwide
- Covered comprehensive full-stack curriculum: HTML5, CSS3, JavaScript (ES6+), React, Node.js, SQL, PHP, TypeScript, GraphQL, CI/CD, and microservices
- Content translated into multiple languages with international distribution across North America, Europe, and Asia
- Presented at educational technology conferences on modern web development pedagogy

Associate Professor & Department Chair

Mount Royal University

Aug 2007 – Dec 2016

Calgary

- Led Computer Science & Information Systems department through university transition and subsequent merger with Mathematics department
- Championed inaugural department chair role, building collaborative culture and driving 50% enrollment growth through industry-aligned curriculum
- Developed municipal transit tracking web application (PHP/MySQL) handling 50k+ daily queries, serving 100k+ transit riders
- Advanced computer literacy initiatives across non-CS disciplines, expanding technology education reach
- Taught diverse course portfolio including software engineering, web development, mobile app development, and introductory programming
- Elected Speaker of General Faculties Council (2014-2015), presiding over university's governing body

Earlier Career Highlights

Software Developer

Clickspace Interactive

Mar 2007 – Jul 2007

Calgary, Alberta

- Developed web applications for e-commerce and insurance claim management using **Ruby on Rails** and **JSP**
- Contributed to agile development environment with cross-functional designer and developer collaboration

Software Developer

Premier GPS

Nov 2006 – Mar 2007

Calgary, Alberta

- Established software development infrastructure from scratch for GPS-to-web application integration
- Designed redundant server systems for emergency signal relay, deployed with HAWKS air ambulance

Web Architect

Alberta Securities Commission

Jul 2006 – Oct 2006

Calgary, Alberta

- Eliminated website downtime (from several minutes/hour to zero) through CMS optimization
- Improved publication workflows including staging-to-production environments and web services integration

Instructor

University of Calgary, Faculty of Continuing Education

Sep 2006 – Nov 2006

Calgary, Alberta

- Taught introductory web design to adult learners with consistently positive student evaluations

Faculty Web Technical Administrator

University of Calgary, Faculty of Social Sciences

Nov 2005 – Jul 2006

Calgary, Alberta

- Rescued compromised infrastructure, securing file systems and deploying enterprise CMS within one week
- Administered redundant web/mail server cluster supporting multiple departmental and research websites
- University-wide CMS selection contribution remains in production today

Team Leader, Software Engineer

PlayStarMusic Corp

Apr 2005 – Nov 2005

Calgary, Alberta

- Built technical infrastructure and team from ground up for venture-backed startup
- Wore multiple hats: system administrator, programmer, designer, team leader, business analyst

Software Engineer

Codefast.com

Dec 2004 – Mar 2005

Calgary, Alberta

- Developed universal compilers in C++, adhering to strict coding standards and requirements

Education

- **M.Sc. Computer Science** – University of Calgary 2004
Thesis: Multi-Agent Modelling and Analysis of Pedestrian and Vehicular Traffic
- **B.Sc. Computer Science** (Minor: Pure Mathematics) – University of Calgary 2002

Selected Publications

- Connolly, R., **Hoar, R.** "Fundamentals of Web Development" (3rd Ed.), Pearson, 2023
- **Hoar, R.**, Connolly, R. "The Garden of Earthly Delights: Constructing and Revising a Web Development Textbook", WWW Conference, 2016
- **Hoar, R.** "A Personalized Web Based Public Transit Information System", IEEE ITSC, 2010
- **Hoar, R.** "Generally Educated In The 21st Century: Computer Literacy In Undergraduate Curriculum", WCCCE, 2014

Honors & Awards

- Nominee, Distinguished Administrator/Manager Award, Mount Royal University 2015-16
- Petro-Canada Young Innovator Award 2008
- Best Student Paper Award, Australian Conference on Artificial Life 2003
- Graduate Student Scholarship, Alberta Learning 2004

Leadership & Service

- Chair, Hanna Public Library Board 2022-2024
- Speaker, General Faculties Council, Mount Royal University 2014-2015
- Secretary, Canadian Association for Computer Science (CACS/AIC) 2011-2016
- Member, Calgary Transit Customer Advisory Group 2010-2015
- Volunteer, ExploreIT & ASTEP Summer Camps (promoting STEM to youth) 2008-2016

Technical Skills

Languages: C++, C#, JavaScript/TypeScript, Python, Swift, Kotlin, PHP, SQL

Frameworks/Platforms: Unreal Engine, Unity, React, Node.js, Ruby on Rails

Tools & Systems: Git, CI/CD, Docker, AWS, MCP, GraphQL, RESTful APIs

Specialties: Test Automation, Game Development, Mobile Development, Developer Tooling, Technical Writing